

**INTRODUCTION TO INDUSTRIAL ENGINEERING - INDR. 100
2025 SPRING**

Class Meeting Location	SCI Z24
Class Meeting Times	Tu,Th, 10:00-11:10
Instructor	CEYDA OĞUZ
Office Hours	Tue 13:15-14:15 or by appointment
Office Location	ENG 223
Office Phone	1793
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Teaching Assistant	Office Hours	Office Location	Email address
Damla Benli	Fri 10:00-11:00	SNA-Z05	dbenli23@ku.edu.tr
Student Teaching Assistant			
Ceren Gülüş	Mon, Wed 11:20-12:20	ENG 216	cgulus25@ku.edu.tr
KOLT Tutor			
Sude Arpat	Mon 17:30-18:45	KOLT	sarpat24@ku.edu.tr
	Tue 11:30-12:45		
	Thu 14:30-15:45		

Course Description

Introduction to industrial engineering concepts. Fundamentals of systems analysis and modeling. Basics of production and service systems.

Course Objectives

The objective of this course is to introduce the profession of industrial engineering and our academic curriculum to freshmen students. Fundamentals of systems analysis and modeling and the basics of production and service systems will be presented. Several industrial engineering topics with their applications in real life systems, modeling of real-life problems and the solution techniques of these models will be introduced.

Learning Outcomes

The students are expected to understand the profession of industrial engineering and its relation with the production and service systems. The students will acquire the fundamentals of systems thinking, analysis and modeling of real-life systems and the basic solution techniques to these models.

Teaching Methods

The objective of this course is to introduce the profession of industrial engineering and our academic curriculum to freshmen students. The course aims to fulfill this task by presenting a compendium of current topics and applications of industrial engineering via lectures and videos. Course material including syllabus and lecture notes, will be at LearnHub.

Textbook

- Turner, W.C., Mize J.H, Case K.E, Nazemetz J.W., Introduction to Industrial and Systems Engineering, 3rd ed., Prentice Hall. (recommended, optional)

Course topics (tentative)

- Introduction to Industrial Engineering (2 weeks)
- Overview of areas in Industrial Engineering (1-3 weeks each)
 - Production and Service Systems
 - Modeling Concepts and Introduction to Linear Programming
 - Introduction to Logistics and Supply Chain Management
 - Network Models and Transportation Problems
 - Probability

Assessment Methods

Type	Final Grade %
Final Exam (comprehensive)	40
Midterm Exam	40
Homeworks	15
Online Modules	5
Total	100

Course Policies:

1. Makeup Exams: The midterm and final exam dates will be announced by the University. Makeup exam for midterm will be given within one week of the announced date. Makeup exam for final will be announced by the University. The student should inform the instructor as early as possible if s/he will not attend the exams. To be eligible for the makeup exams, the students' official medical report issued by the University should reach the instructor at the latest the next day of the announced exam date.
2. Remedial Exam: Since this course is offered in both semesters, there will be NO remedial exam. If you have questions, please see the University regulations regarding this issue.
3. Homeworks: There will be two homework assignments to be graded. These homework assignments are individual assignments and are designed to help you to get ready for the exams. You can discuss the topics related to the homework questions with your classmates and/or TAs however each student should solve the questions given in the homework by herself/himself. Checking the correctness of your solutions with your classmates before submission and correcting them based on this comparison are also forbidden. Please see the below part on **Academic Dishonesty**.
4. Late submission policy: Late homework submission will have the following penalty:
If a homework assignment is submitted within k days later than the due date, the assignment will be evaluated out of $(100-10k)\%$, where k can be at most 3.
Please observe the deadlines and do not wait for the last moment. Technical problems always happen and they are not excuses that I will accept.
5. Study Questions: Study questions will be given as we progress through the topics. Students can get help from TA for these questions however each student should solve the questions before seeking any help.
6. Attendance: This is NOT a compulsory assessment component, but attendance will be recorded. Attending lectures will help you to better understand the content, and to interact with the instructor and fellow students.
7. Office Hours: Office hours should be used for asking questions related to the course. If there is a need, TAs and the instructor can be contacted via e-mail outside these office hours and students should expect a reply within 24 hours. Please do not flood us with e-mails before 24 hours of your first e-mail.
8. LearnHub: LearnHub will be used as the official site of the course. Lecture notes, online modules, study questions and announcements will be posted at LearnHub. Students should check LearnHub regularly.
9. Recording of Lectures: The lectures will be held face-to-face and there will be no hybrid teaching. Hence, recorded lecture videos will NOT be provided.
10. Student code of conduct: Students should obey the University rules and regulations as set at <https://apdd.ku.edu.tr/en/academic-policies/student-code-of-conduct/>
11. AI Policy: Students are free to use any tools to support their learning, including AI-powered queries and chatbots (such as ChatGPT). You should always keep in mind that such tools are most useful when you have good background knowledge about the topic of interest and understand what the current AI procedures can and cannot do. Like any new and sophisticated tools, these tools should be used responsibly. If you use these tools for any task related to the class, you are required to acknowledge them openly and clearly. Failing to report the use of these tools in any course task will be considered cheating, and required disciplinary action will be taken.

Online Modules

There will be several online modules that you need to complete within this course. After you watch the video for each online module, you will take the online exam. You will get either a “pass” or “fail” from these exams. These modules will contribute 5% of your final grade only if you will pass both online modules. The videos will be uploaded to the LearnHub under this course heading.

- **Module 1: Engineering Ethics:**
 - **Module 1a: Introduction to Ethics** consists of two video recordings, which include an introduction to the subject of ethics and a comparison of the concepts of ethics and morality, and a test exam.
 - **Module 1b: Ethics from the perspective of engineering** consists of a video recording containing the code of ethics on a departmental basis and a test exam.
- **Module 2: Engineering Standards** includes a video recording of introductory lectures on engineering standards and a test exam.

Links

- <http://www.iienet.org> (IIE) Institute of Industrial Engineers
- <http://www.informs.org> (INFORMS) Institute for Operations Research and Management Sciences
- <http://www.euro-online.org> (EURO) Association of European Operational Research Societies
- <http://www.ifors.org> (IFORS) International Federation of Operational Research Societies
- <http://www.yad.org.tr> (YAD) Yöneylem Araştırması Derneği

Classroom Code of Conduct: (taken from <https://apdd.ku.edu.tr/en/academic-policies/student-code-of-conduct/>)

Students at Koç University are required to adhere to classroom code of conduct and to refrain from all forms of unacceptable behavior during lectures. The activities which are prohibited in class include and are not limited to:

- Engaging in side conversations.
- Using cell phones and other electronic devices, unless permitted by the instructor.
- Using laptops for purposes that are not course related.
- Arriving late or leaving early without the prior permission of the instructor.
- Reading material, e.g., magazines, newspapers, novels etc., that are not course related.
- Working on personal activities or the assignments of other courses.
- Interrupting the professor or other students (students should request permission from the instructor before asking questions or making comments).
- Trashing the classroom.
- Eating and drinking, except water.
- Students should be aware of **Koç University Policy Statement Regarding the Use of Artificial Intelligence Tools (ChatGPT or other similar tools)**

Failure to comply with the Classroom Code of Conduct may result in dismissal from class and disciplinary action.

Academic Dishonesty: Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct. Actions considered as academic dishonesty at Koç University include but are not limited to cheating, plagiarism, multiple submissions and collusion. Copying from others or providing answers or information, written or oral, to others in any assessment component is cheating. Copying from another student's paper or from another text or internet site without written acknowledgment is plagiarism. Resubmitting one's own work in another course or project is a form of multiple submission. Unauthorized help from another person or having someone else write one's paper or assignment is collusion. Cheating, plagiarism, multiple submission and collusion are serious offenses resulting in an F grade and disciplinary action. Facilitating academic dishonesty is equivalent to above actions.